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ceph



ROOK

Storage 101

Rook and Ceph



OPEN INFRASTRUCTURE
SUMMIT

Denver, CO | April 29-May 1, 2019

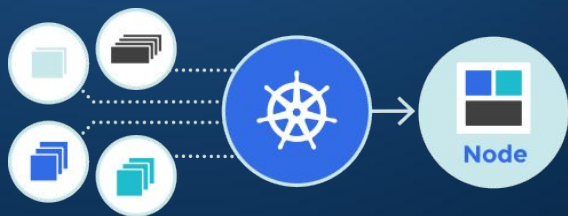


WHAT IS THIS TALK ABOUT?

STORAGE CONTAINER ORCHESTRATION WITH KUBERNETES



Kubernetes (K8s) is an open-source system for automating deployment, scaling, and management of containerized applications.





Ceph is an open source distributed storage software-defined solution that allows you to consume your data through several interfaces such as object, block and file.



Ceph Architecture

APP



HOST/VM



CLIENT



RGW

A web services gateway for object storage, compatible with S3 and Swift

RBD

A reliable, fully-distributed block device with cloud platform integration

CEPHFS

A distributed file system with POSIX semantics and scale-out metadata management

LIBRADOS

A library allowing apps to directly access RADOS (C, C++, Java, Python, Ruby, PHP)

RADOS

A software-based, reliable, autonomous, distributed object store comprised of self-healing, self-managing, intelligent storage nodes and lightweight monitors



WHY IS STORAGE HARD?

STORAGE IN KUBERNETES CONTAINER ORCHESTRATION



STORAGE IN KUBERNETES

- K8s abstracts away the infrastructure it manages
- Dynamic environment
 - Balancing load
 - Rebuilding pods (healing)
- Ephemeral storage design



Terminology

- **CRD:** Custom Resource Definition; Schema Extension to Kubernetes API
- **CR:** Custom Resource; One record/instance/object, conforming to a CRD
- **OPERATOR:** Daemon that watches for changes to resources
- **STORAGE CLASS:** “class” of storage service
- **PVC:** Persistent Volume Claim, attach persistent storage to a pod
- **POD:** a group of one or more containers managed by Kubernetes

Cloud-native storage landscape

Runtime



Arrikto



DATARA

DELL EMC

DIAMANTI

elastifile



HATCHWAY



Hewlett Packard
Enterprise

INFINIDAT



LeoFS

LONGHORN



MooseFS



OpenEBS
by MAYADATA

openIO

openSDS



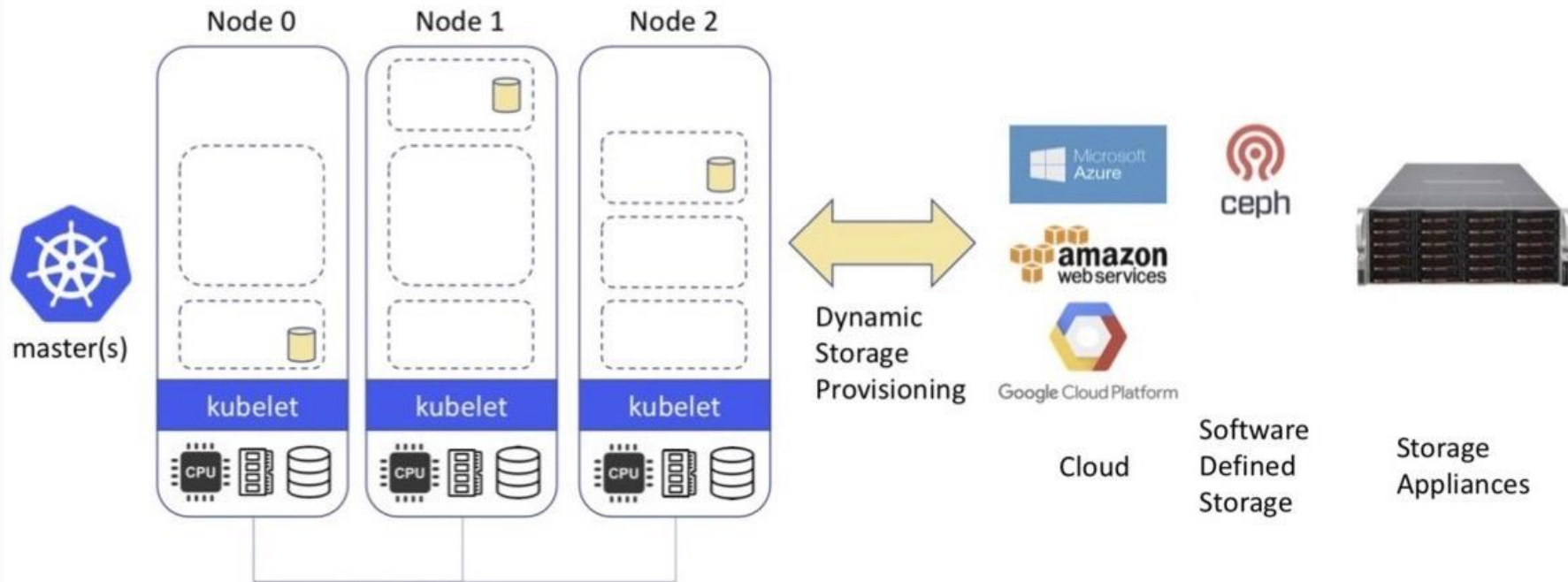
STORAGEOS



TRITON
Object
Storage



Storage for Kubernetes



Volume plugins allow external storage solutions to provide storage to your apps



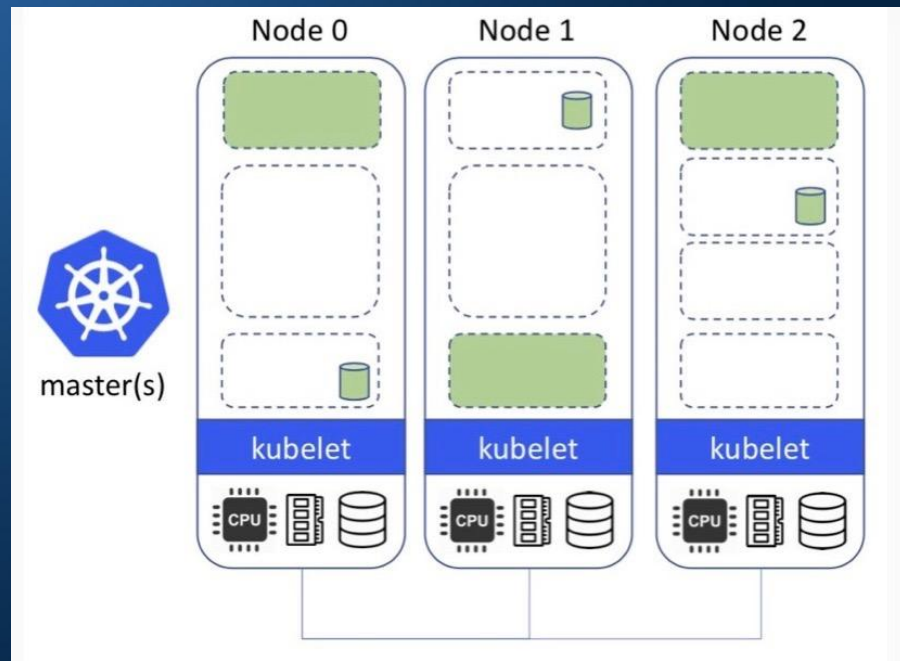
Limitations

- Not portable: requires these services to be accessible
- Deployment burden of external solutions
- Vendor lock-in due to using provider managed services

Storage ON Kubernetes

Kubernetes can manage our storage solution

- Highly portable applications (including storage dependencies)
- Dedicated K8s storage cluster also possible





Portable Abstractions

Concept that describes the general data need of your application

- Database, message queue, cache, object store, etc.
- Works across environments and providers
 - RDS in Amazon
 - Cloud SQL in Google
- Choose the best environment for your code
 - cost, features, resiliency, compliance
- Goal: Hybrid and Multi Cloud Storage abstraction



The Container Storage Interface

(CSI)





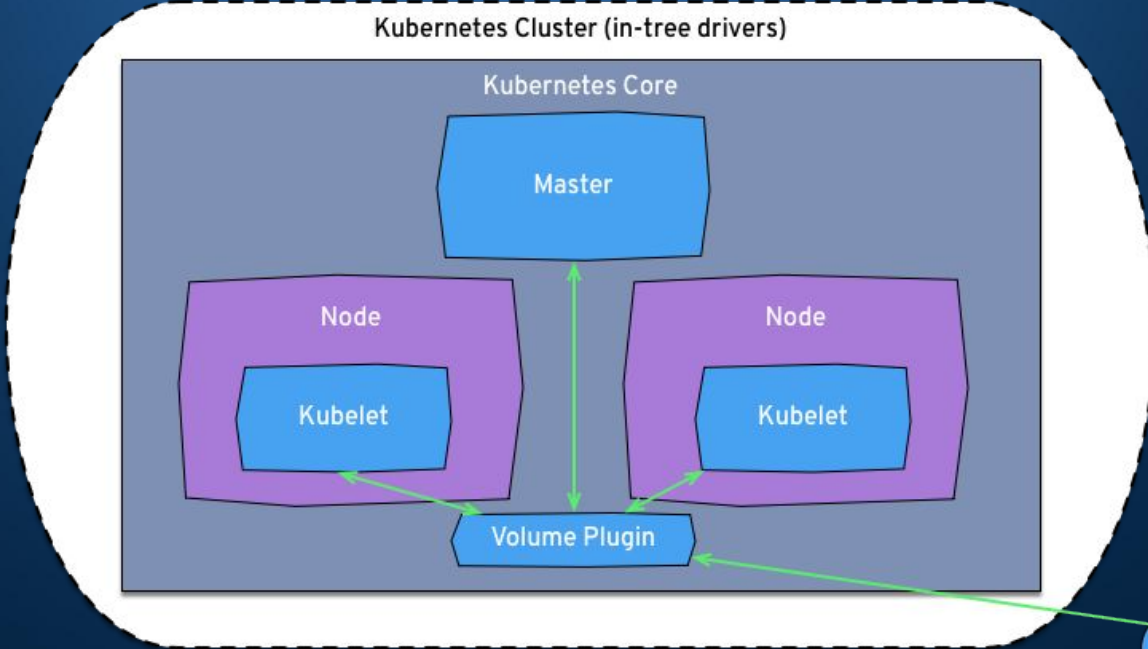
CSI



Container Storage Interface

- Prior to CSI, it was challenging to add support for new volume plugins to Kubernetes.
- Volume plugins were “in-tree”, third-party storage code caused reliability and security issues in core Kubernetes binaries
- With the introduction of CSI, storage can now be treated as another workload to be containerized and deployed on a Kubernetes cluster.
- Using CSI, third-party storage providers can write and deploy plugins exposing new storage systems in Kubernetes without touching the core Kubernetes code.

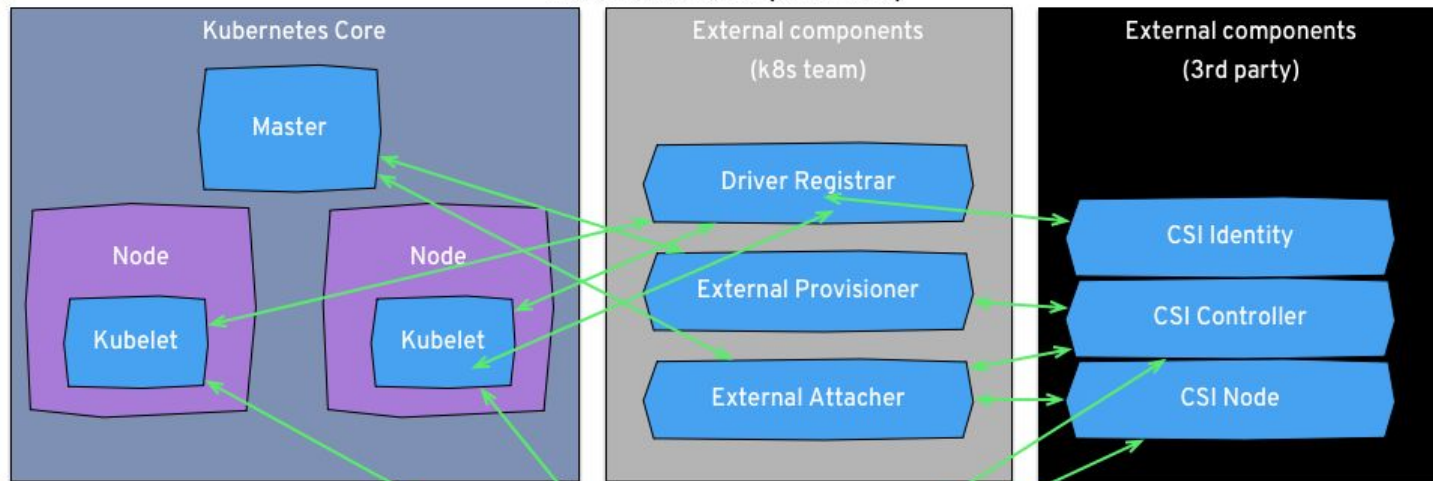
CSI



CSI

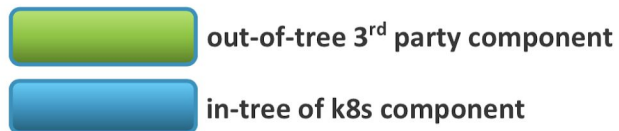
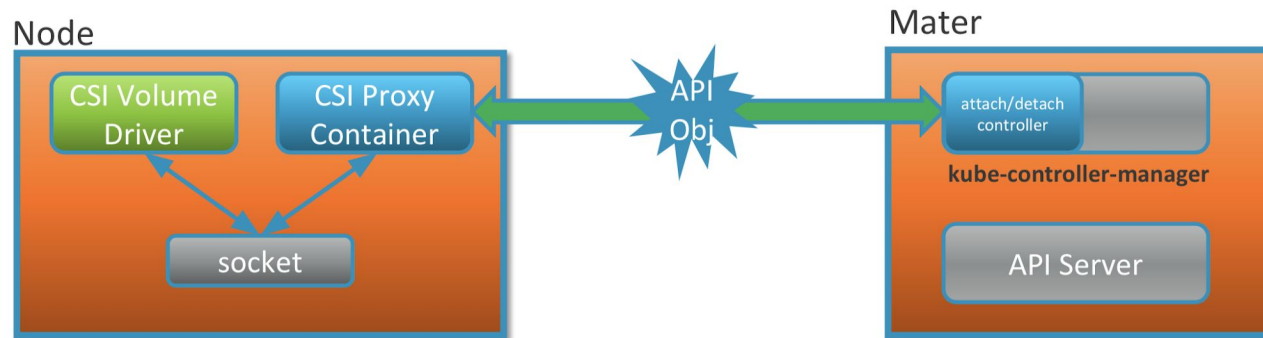


Kubernetes Cluster (CSI drivers)





- A new in-tree CSI Volume plugin(K8s) + out-of-tree CSI Volume driver (3rd party)
- Communication channel via a Unix Domain Socket(UDS) created by 3rd Volume Driver



The socket file also called a 'EndPoint' in form of like:
/var/lib/kubelet/plugins/rook-ceph/csi.sock



- Ceph CSI plugin allows dynamically provisioning Ceph volumes and attaching them to workloads.
- Relies on Kubernetes CSI spec (v3.0 and v1.0)
- Integrated in Rook 1.0
 - <https://github.com/ceph/ceph-csi/>

Storage access modes:

- RWO - ReadWriteOnce: the volume can be mounted as read-write by a single node
- ROX - ReadOnlyMany: the volume can be mounted read-only by many nodes
- RWX - ReadWriteMany: the volume can be mounted as read-write by many nodes



Support Matrix

VOLUME TYPE	FEATURES	CSI DRIVER VERSION
File mode, sharable or RWX Volume(CephFS)	Dynamically provision, de-provision volume	v0.3.0
	Creating and deleting snapshot	-
	Provision volumes from snapshot	-
	Provision volumes from another Volume	-
	Resize volumes	-
Block mode, sharable or RWX volumes(RBD) File/Block mode single-consumer or RWO volumes(RBD)	Dynamically provision, de-provision volume	v0.3.0,v1.0.0
	Creating and deleting snapshot	v0.3.0,v1.0.0
	Provision volumes from snapshot	v1.0.0
	Provision volumes from another Volume	-
	Resize volumes	-



ROOK

PROVIDE BEST CEPH STORAGE EXPERIENCE IN KUBERNETES



ROOK



ROOK + CEPH



- Rook is bringing Ceph and Kubernetes together
- It is one of the most popular cloud-native storage solutions
- With more than 5000 Github stars, 21M docker pools and 140+ contributors.
- Accepted as the CNCF's first storage project
- Rook has recently reached incubation stage



Storage Framework

Cloud-Native Storage Orchestrator

- Extends Kubernetes with custom types and controllers
- Automate deployment, bootstrapping, configuration, provisioning, scaling, upgrading, migration, disaster recovery, monitoring, and resource management
- Framework for many storage providers and solutions
- Open Source (Apache 2.0)
- Hosted by the Cloud-Native Computing Foundation





Storage Framework

- Orchestration
 - Automated deployment, bootstrapping, configuration, provisioning, scaling, upgrading, migration, disaster recovery, monitoring, and resource management
- Kubernetes Volume Plugins: Attaching and mounting RBD/CephFS
 - In-tree drivers (legacy)
 - Flex Volume Driver (soon to be deprecated)
 - Ceph-CSI (Container Storage Interface)



Advantages

THE BENEFIT OF CEPH+ROOK

- Ceph automates the internal storage management
- Rook automates the user-facing operations
- Rook makes storage consumption from Kubernetes transparent

SCALE

- Horizontally (nodes) and vertically (user volumes), storage scale (Ceph)
- Fast, automatic provisioning
- Hides ceph.conf settings from end user (could be any storage source)



Why Rook?

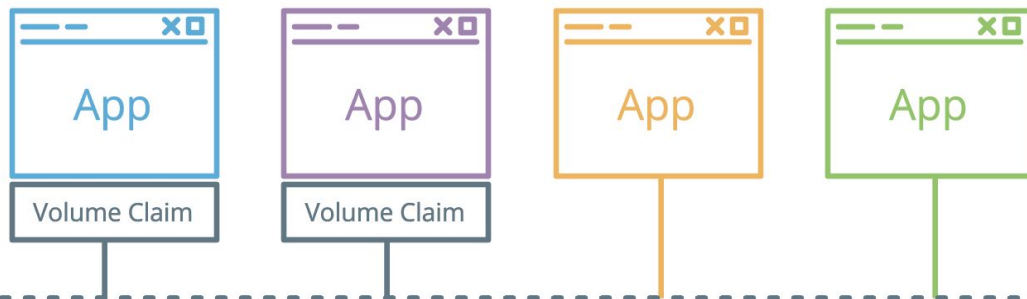
Kubernetes Storage Challenges

- Reliance on external storage
 - Requires these services to be accessible
 - Deployment burden
- Reliance on cloud provider managed services
 - Vendor lock-in
- Day 2 operations - who is managing the storage?

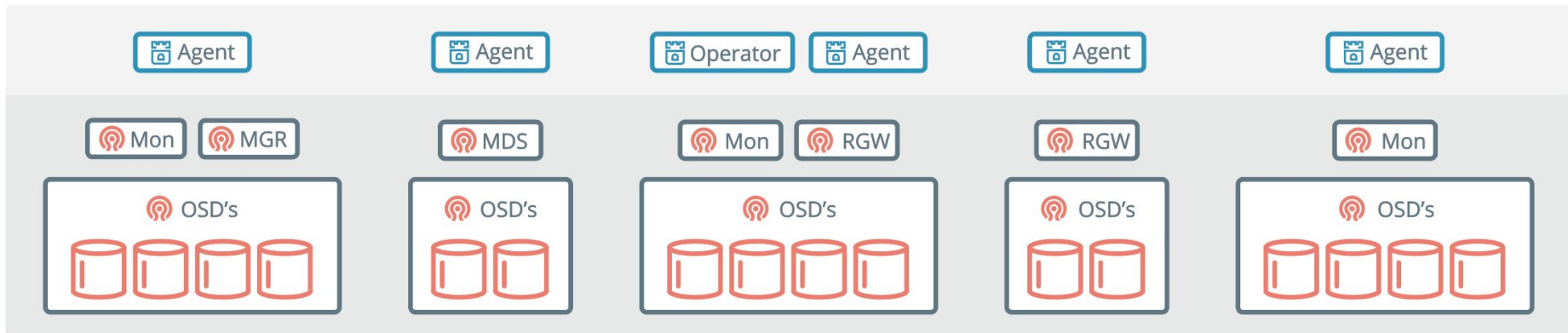


ARCHITECTURE

High-level Architecture



ROK pods





Rook Operator

- Implements the **Operator Pattern** for Ceph
 - Existed before the operator-sdk or kubebuilder
- User defines *desired state* for the storage cluster
- Operator:
 - Observes - Watch for changes in state and health
 - Analyzes - Determine differences to apply
 - Acts - Apply changes to the cluster



Cluster CR

```
apiVersion: ceph.rook.io/v1
kind: CephCluster
metadata:
  name: rook-ceph
  namespace: rook-ceph
spec:
  cephVersion:
    image: ceph/ceph:v14.2
  mon:
    count: 3
  dashboard:
    enabled: true
  storage:
    useAllNodes: true
    useAllDevices: true
```



Rook-Ceph Operator

- Automates configuration of the following Ceph daemons:
 - MON: Create mons and ensure they are in quorum
 - MGR: Start the daemons
 - OSD: Provision devices with ceph-volume and start daemons
 - RGW: Create the object store and start the RGW daemons
 - MDS: Create the CephFS and start the MDS daemon
 - NFS: Start the NFS Ganesha daemons
 - RBD Mirroring: Start the daemons

No iscsi-gw yet.



Available CRDs

Rook's Custom Resource Definitions (CRDs):

- CephCluster: represents a Ceph Cluster
- CephBlockPool: represents a Ceph Block Pool
- CephFilesystem: represents a Ceph Filesystem interface
- CephNFS: represents a Ceph NFS interface.
- CephObjectStore: represents a Ceph Object Store.
- CephObjectStoreUser: represents a Ceph Object Store User.



FUTURE OF ROOK

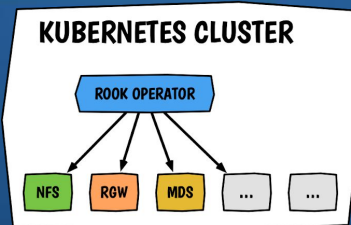


External Cluster

- Ability to **consume** existing Ceph storage cluster that were not deployed with Rook
- Rook does **not** manage the cluster
- Bootstrap stateless daemons in Kubernetes but leave the rest in place on the existing cluster
- Different Storage Classes for certain clusters

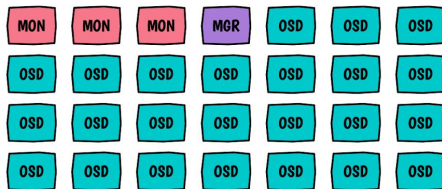
External Cluster

KUBERNETES CLUSTER



```
kind: CephCluster
spec:
  external: true
  fsid: 1a394bf1-b42b-4dc9-94e7-85848fc750fd
  monitors: 192.168.0.1,192.168.0.2,192.168.0.3
  adminSecret: QVFEMGdwdGNENFNpQmhBQXZYVjJJY2lTK1p4eG9vN3E0Wn1vUWc9PQo=
  name: rook-ceph-external
```

EXTRENAL CEPH CLUSTER





Incoming changes

- Auto-scale when plugging a new disk
- Watch for new storage node and increase capacity automatically
- Upgrade mechanism enhancement
- Expose more CR's details (Ceph health)
- More control over logging (enable/disable on the fly)
- Better maintenance mode
- Better resources control (requests and limits)



Future Work

- Integration with Multus
 - Attach multiple physical interfaces to a pod
 - Removes the need of host Networking
 - More secure, more control
- Cloud block provisioner
 - Relies on Kubernetes block PVC
 - Allows smoother run Cloud platforms like AWS/GKE/AKS
- Bucket Storage Class for object storage
 - `ObjectBucketClaim` / `ObjectBucket`



Give it a try!

- Download minikube

```
minikube start
```

```
git clone https://github.com/rook/rook
```

```
cd cluster/examples/kubernetes/ceph
```

```
kubectl create -f common.yaml operator.yaml
```

```
kubectl create -f cluster.yaml
```



Getting Started with Rook

- <https://rook.io>
 - Read the docs... seriously
 - Github repo: <https://github.com/rook/rook>
 - Design docs:
<https://github.com/rook/rook/tree/master/design>
- Slack
<https://rook-io.slack.com/>



THANK YOU



@0xF2
@leseb_
@SeanCohen_RH

The Red Hat Storage logo, featuring the text 'RED HAT' in a smaller font above the word 'STORAGE' in a larger, bold font, both in white on a red rectangular background.

RED HAT
STORAGE